

Since the mid-1980s, several *Panthera* species became subjects of genetic research, mostly using blood samples of captive individuals. Study results indicate that many of the lion and leopard subspecies are questionable because of insufficient genetic distinction between them.^{[28][29]} Subsequently, it was proposed to group all African leopard populations to *P. p. pardus* and retain eight subspecific names for Asian leopard populations.^[30] Results of genetic analysis indicate that the snow leopard (formerly *Uncia uncia*) also belongs to the genus *Panthera* (*P. uncia*), a classification that was accepted by IUCN Red List assessors in 2008.^{[9][31]}

Based on genetic research, it was suggested to group all living sub-Saharan lion populations into *P. l. leo*.^[32] Results of phylogeographic studies indicate that the Western and Central African lion populations are more closely related to those in India and form a different clade than lion populations in Southern and East Africa; southeastern Ethiopia is an admixture region between North African and East African lion populations.^{[33][34]}

Black panthers do not form a distinct species, but are melanistic specimens of the genus, most often encountered in the leopard and jaguar.^{[35][36]}

Contemporary species

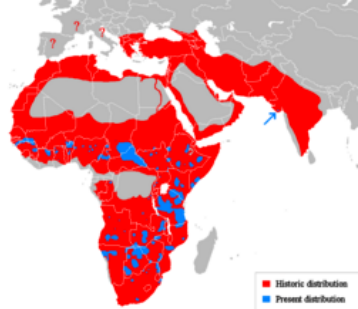
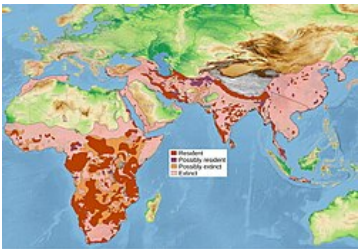
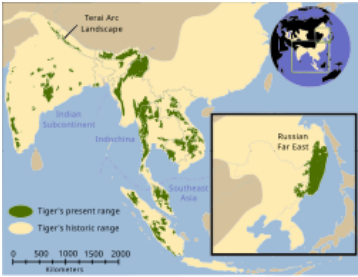
The following list of the genus *Panthera* is based on the taxonomic assessment in *Mammal Species of the World* and reflects the taxonomy revised in 2017 by the Cat Classification Task Force of the Cat Specialist Group.^{[2][37]}

- *Panthera uncia* (Schreber, 1775)
- †*Panthera youngi* Pei, 1934
- †*Panthera zdanskyi* Mazák, Christiansen & Kitchener, 2011

Synonyms^[2]

List

- *Jaguaris* Severtzov, 1858
- *Leo* Frisch, 1775
- *Leonina* Greve, 1894
- *Leoninae* Wagner, 1841
- *Pardotigris* Kretzoi, 1929
- *Pardus* Fitzinger, 1868
- *Tigrina* Greve, 1894
- *Tigrinae* Wagner, 1841
- *Tigris* Gray, 1843
- *Tigris* Frisch, 1775

Species	Subspecies	IUCN Red List status and distribution
<u>Lion</u> <i>P. leo</i> (Linnaeus, 1758)^[38] 	<i>P. l. leo</i> (Linnaeus, 1758)^[38] including: †<u>Barbary lion</u> <i>P. l. leo sensu stricto</i> <u>Asiatic lion</u> syn. <i>P. l. persica</i> (Johann Nepomuk Meyer, 1826)^[39] <i>P. l. melanochaita</i> (Smith, 1842)^[40] including: †<u>Cape lion</u> <i>P. l. melanochaita sensu stricto</i>^[41]	VU^[42] 
<u>Jaguar</u> <i>P. onca</i> (Linnaeus, 1758)^[38] 	Monotypic^{[43][37]}	NT^[44] 
<u>Leopard</u> <i>P. pardus</i> (Linnaeus, 1758)^[38] 	<u>African leopard</u> <i>P. p. pardus</i> (Linnaeus, 1758)^[38] <u>Indian leopard</u> <i>P. p. fusca</i> (Meyer, 1794)^[45] <u>Javan leopard</u> <i>P. p. melas</i> (G. Cuvier, 1809)^[46] <u>Arabian leopard</u> <i>P. p. nimr</i> (Hemprich and Ehrenberg), 1833^[47] <i>P. p. tulliana</i> (Valenciennes, 1856),^[48] syn. <i>P. p. ciscaucasica</i> (Satunin, 1914),^[49] <i>P. p. saxicolor</i> Pocock, 1927^[50] <u>Amur leopard</u> <i>P. p. orientalis</i> (Schlegel, 1857),^[51] syn. <i>P. p. japonensis</i> (Gray, 1862)^[52] <u>Indochinese leopard</u> <i>P. p. delacouri</i> Pocock, 1930^[53] <u>Sri Lankan leopard</u> <i>P. p. kotiya</i> Deraniyagala, 1956^[54]	VU^[55] 
<u>Tiger</u> <i>P. tigris</i> (Linnaeus, 1758)^[38] 	<i>P. t. tigris</i> (Linnaeus, 1758) including: <u>Bengal tiger</u> <i>P. t. tigris sensu stricto</i> †<u>Caspian tiger</u> syn. <i>P. t. virgata</i> (Illiger, 1815)^[56] <u>Siberian tiger</u> syn. <i>P. t. altaica</i> (Temminck, 1844)^[57] <u>South China tiger</u> syn. <i>P. t. amoyensis</i> (Hilzheimer, 1905)^[58] <u>Indochinese tiger</u> syn. <i>P. t. corbetti</i> Mazák, 1968^[59] <u>Malayan tiger</u> syn. <i>P. t. jacksoni</i> Luo et al., 2004^[60] <u>Sunda Island tiger</u> <i>P. t. sondaica</i> Temminck, 1844)^[57] including †<u>Javan tiger</u> <i>P. t. sondaica sensu stricto</i> <u>Sumatran tiger</u> syn. <i>P. t. sumatrae</i> (Temminck, 1844)^[57] †<u>Bali tiger</u> syn. <i>P. t. balica</i> Schwarz, 1912)^[61]	EN^[62] 

Snow leopard *P. uncia*^[37] (Schreber, 1775)^[63]



Monotypic^[37]

vu^[31]



Extinct species and subspecies

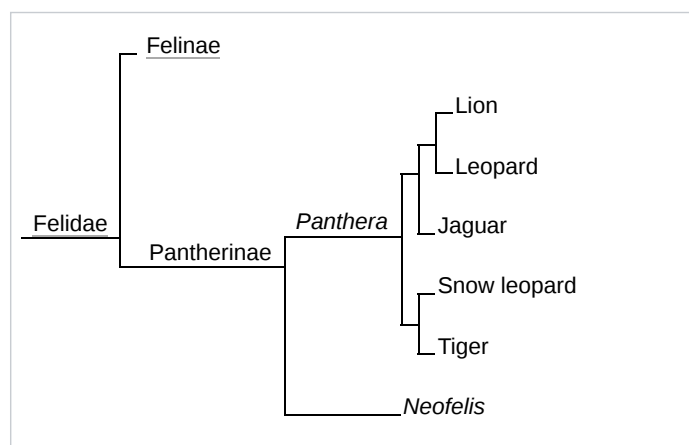
Species and subspecies	Fossil records	Notes	Images
<i>Panthera atrox</i>	North America, 0.13 to 0.013 MYA, with dubious remains in South America. ^[64]	Commonly known as the American lion, <i>P. atrox</i> is thought to have descended from a basal <i>P. spelaea</i> cave lion population isolated south of the Cordilleran Ice Sheet, and then established a mitochondrial sister clade circa 200,000 BP. ^[65] It was sometimes considered a subspecies either under the nomenclature of <i>P. leo</i> ^[65] or <i>P. spelaea</i> . ^[66] One of the largest <i>Panthera</i> species. ^[67] Became extinct around 13,000-12,000 years ago. ^[68]	
<i>Panthera balamoides</i> ^[69]	Mexico, ~0.13 MYA	Dubious, other authors suggest that the remains are actually of the extinct bear <i>Arctotherium</i> instead. ^[70]	
<i>Panthera fossilis</i> ^[71]	Europe and Asia, 0.68 to 0.25 MYA	Extinct species of lion known from the Middle Pleistocene of Europe and Asia. One of the largest known species of <i>Panthera</i> . Considered to be the ancestor of <i>P. spelaea</i> . ^[72]	
<i>Panthera gombaszogensis</i>	Europe, possibly Asia and Africa, 2.0 to 0.35 MYA	Ranged across Europe, as well as possibly Asia and Africa from around 2 million to 350,000 years ago. ^[73] Often suggested to be the ancestor of the living jaguar (<i>Panthera onca</i>), and sometimes referred to as the "European jaguar". <i>Panthera schreuderi</i> and <i>Panthera toscana</i> are considered junior synonyms of <i>P. gombaszogensis</i> . It is occasionally classified as a subspecies of <i>P. onca</i> . ^{[74][75]}	
<i>Panthera palaeosinensis</i>	Northern China, ~3 MYA	Initially thought to be an ancestral tiger species, but several scientists place it close to the base of the genus <i>Panthera</i> . ^[1] At least three recent studies considered <i>Panthera zdanskyi</i> likely to be a synonym of <i>P. palaeosinensis</i> . ^{[76][77][78]}	
<i>Panthera principalis</i>	Tanzania, ~3.7 MYA	Described in 2023. ^[13]	
<i>Panthera shawi</i>	Laetoli site in Tanzania, ~3 MYA	A leopard-like cat ^[79]	
<i>Panthera spelaea</i>	Much of Eurasia, 0.6 to 0.013 MYA ^[80]	Commonly known as the cave lion or steppe lion. Originally <i>spelaea</i> was classified as a subspecies of the extant lion <i>P. leo</i> . ^[81] Results of recent genetic studies indicate that it belongs to a distinct species, namely <i>P. spelaea</i> that is most closely related to the modern lion among living <i>Panthera</i> species. ^{[82][83]} Other genetic results indicate that <i>P. fossilis</i> also warrants status as a species. ^{[84][85]} It became extinct around 14,500-14,000 years ago. ^[86]	
<i>Panthera youngi</i> ^[87]	China, ~0.35 MYA		

<i>Panthera zdanskyi</i>	Gansu province of northwestern China, 2.55 to 2.16 MYA	It was initially considered to be a close relative of the tiger. ^[1] But it is possibly synonymous with <i>P. palaeosinensis</i> . ^{[13][88]}	
<i>Panthera leo sinhaleyus</i>	Sri Lanka	This lion subspecies was described on the basis of two teeth. ^[89]	
<i>Panthera onca augusta</i> ^[90]	North America	May have lived in temperate forests across North America ^[91]	
<i>Panthera onca mesembrina</i> ^[92]	South America	May have lived in grasslands in South America, unlike the modern jaguar	
<i>Panthera pardus spelaea</i>	Europe	Closely related to Asiatic leopard subspecies, ^[93]	
<i>Panthera tigris acutidens</i>	Much of Asia	Not closely related to modern tiger subspecies ^[94]	
<i>Panthera tigris soloensis</i>	Java, Indonesia	Not closely related to modern tiger subspecies ^[94]	
<i>Panthera tigris trinilensis</i>	Java, Indonesia	Not closely related to modern tiger subspecies ^[94]	

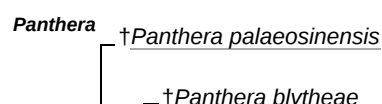
Other, now invalid, species have also been described, such as *Panthera crassidens* from South Africa, which was later found to be based on a mixture of leopard and cheetah fossils.^[95] A "*Panthera dhokpathanensis*" was briefly referenced in 1986 in a report on apparent new carnivorans from the Dhok Patha region in the Siwaliks, but as no description was provided this name is a *nomen nudum*.^[96]

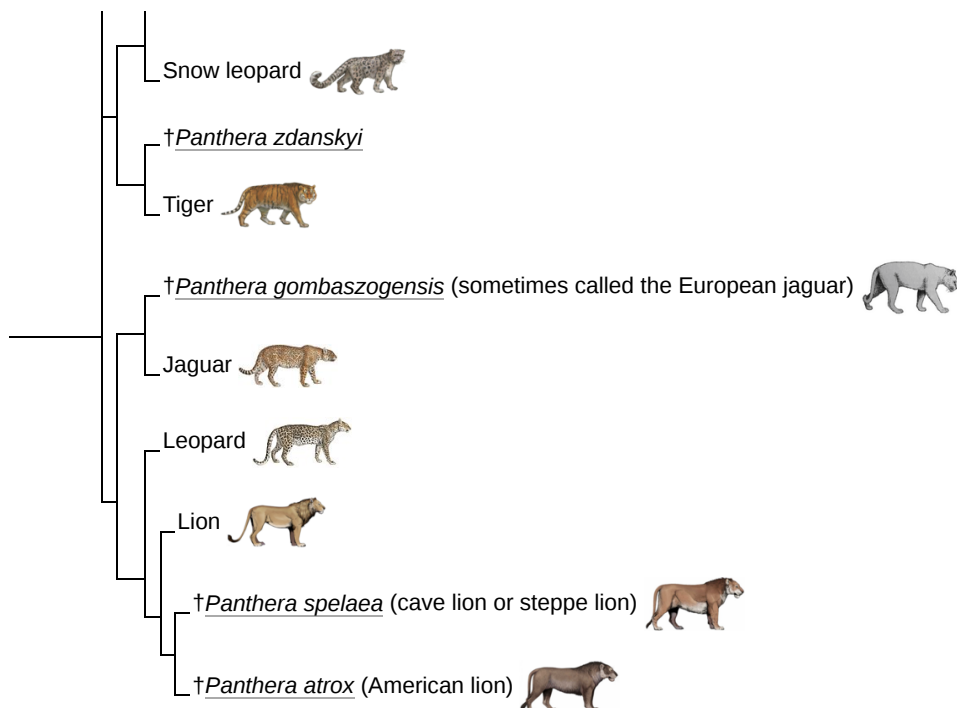
Phylogeny

Results of a 2016 study based on analysis of biparental nuclear genomes suggest the following relationships of living *Panthera* species:^[97]



In 2018, results of a phylogenetic study on living and fossil cats were published. This study was based on the morphological diversity of the mandibles of saber-toothed cats, their speciation and extinction rates.^[99]





Two cladograms proposed for *Panthera*. The upper one is based on phylogenetic studies by Johnson et al. (2006),^[9] and by Werdelin et al. (2010),^[98] The lower cladogram is based on a study by Davis et al. (2010)^[19] and by Mazák et al. (2011).^[1]



Notes

1. /'pænθərə, pæn'ti:rə/ *PAN-thə-rə, pan-THEE-rə*^{[3][4]}

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Further reading

- Turner, A.; Antón, M. (1997). *The Big Cats and Their Fossil Relatives: An Illustrated Guide to Their Evolution and Natural History* (<https://books.google.com/books?id=66mRJSxIAfoC>). Columbia University Press. ISBN 978-0-231-10228-5.

External links

-  Media related to *Panthera* at Wikimedia Commons
- *The Ghostly Origins of the Big Cats* (<https://www.youtube.com/watch?v=oPJnqWke5n8>) (video). PBS Eons. 16 May 2019. Archived (<https://ghostarchive.org/varchive/youtube/20211211/oPJnqWke5n8>) from the original on 11 December 2021 – via YouTube.

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